SYNTHAGENT: ~~A System for~~ Reliable LLM-Based Synthetic Data Generation ~~in~~ for Small ~~Data~~Healthcare Datasets

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***Abstract*—Prediction ~~models in healthcareare often trained on~~ tasks in healthcare, such as diagnosing a disease or anticipating an adverse outcome, often have to be learned from small datasets, ~~in which the number of records is too low for the complexity of the task, for example because a~~ because the disease is rare or because labelling records is expensive. Synthetic data generation can be a solution for such contexts, but established generators, such as oversampling techniques and generative adversarial networks, learn only from the available records~~and can therefore only reproduce~~ , recombining patterns that already exist in the data. Large language models (LLMs) can ~~go beyond these patterns, because they bring~~ address this issue by adding domain knowledge acquired during pre-training~~and do not need to be~~ , without being trained on the target data. When prompted directly, however, LLMs often return fewer records than requested, malformed records or values of the wrong type, so their output cannot be used as a dataset~~without further processing~~. To address this issue, we present SYNTHAGENT, a ~~system that places~~ synthetic data generation system for small healthcare datasets that combines an LLM with a control layer~~between the LLM and the synthetic dataset. The control layer infers the names and types of the columns~~ . Inferring the columns and their types from the real data, the control layer checks every response of the LLM~~against them, asks the model to correct the values that violate them, and keeps requesting~~ , asks for the correction of invalid values and requests new records until the ~~number set by the user~~ configured number is reached, while recording the cost of every request. ~~was evaluated~~ In an evaluation on six public healthcare datasets~~with four LLMs, three established generators and ten classifiers. It completed every synthetic dataset to the requested number of records, which~~ , SYNTHAGENT completed the synthetic datasets to the configured size, which neither direct prompting of the same LLM ~~never achieved~~ nor schema-constrained decoding achieved in any run, at a ~~cost only 1.9% higher. With the strongest LLM, adding its records~~ small additional cost. Added to the training data ~~increased the average F1-score of the classifiersby 1.1 percentage points, the largest~~ of ten classifiers, its records yielded the largest average F1-score gain among the generators compared ~~, and was the only generator whose records never lowered the F1-score of any dataset by more than one point~~and avoided the losses of up to 10.6 points that SMOTE, CTGAN and TVAE caused on individual datasets.

***Index Terms*—small data, synthetic data generation, large language models, ~~tabular data~~ healthcare

I. INTRODUCTION

Many prediction tasks in healthcare, such as diagnosing a disease or anticipating an adverse outcome, suffer from a lack of data. A rare disease may have only a few dozen diagnosed cases, a study conducted in a single healthcare facility is limited to the patients it can recruit, and labelling records often requires expert review. Data sharing between ~~institutions~~ facilities could increase the number of records, but it is frequently restricted by privacy regulations [1]~~. In this paper~~ , so each facility often has to learn from its own data. In our work we refer to such datasets as *small data*: datasets whose number of records is ~~low relative to what the learning task requires, which~~ too low to train a reliable prediction model, a limit that depends on the number of features, the number of classes and the frequency of the outcome [2]. A common way to mitigate small data is synthetic data generation (SDG), which adds artificial records that follow the distribution of the real ones to the training set [3]. This paper focuses on SDG for prediction tasks on small healthcare datasets.

SDG is, however, hardest when data are small. Oversampling techniques, such as the Synthetic Minority Oversampling Technique (SMOTE) [4], and deep generative models, such as the conditional tabular generative adversarial network (CTGAN) and the tabular variational autoencoder (TVAE) [5], learn the distribution from the same few records they are meant to complement~~. They can only recombine patterns already present~~ , recombining patterns that already exist in those records, for example by interpolating between two existing patients~~, and cannot introduce knowledge that the records do not reflect~~ [6]. Large language models (LLMs) offer a ~~different source of information~~new perspective to address such issue. They are pre-trained on large text corpora, which gives them domain knowledge about healthcare variables, such as plausible ranges and typical relations between them, and they can generate new records from a few real examples placed in the prompt, without any training on the target dataset [7], [8]. Their output, however, is free text rather than a dataset: a response may contain fewer records than requested, malformed records, missing or invented columns,

or values of the wrong type (Figure 1)~~, and the cost of each request is usually not recorded~~. Every augmentation method is configured with the number of synthetic records to add, usually expressed as a proportion of the real training set [4], and a generator that silently returns fewer or malformed records changes the training data without the user noticing. In addition, each request to a commercial LLM has a monetary cost, proportional to the number of tokens (word fragments) processed, which is rarely reported. One problem that stands out is therefore how to obtain from an LLM a synthetic dataset ~~that has the~~ with the configured number of records~~requested by the user, conforms to~~, the columns and types of the real data and ~~has~~ a known cost, so that it can be used to improve the ~~performance of~~ classifiers trained on small data. Formally, given the training records D of a small dataset, their columns and types S and the configured number of synthetic records N, a configuration θ of the generator produces a synthetic dataset G_θ. The goal is to find a configuration whose dataset belongs to the set $\mathcal{F}(S, N)$ of datasets with exactly N records that conform to S and that maximizes the gain in classifier performance ΔU, while keeping the distance from the real distribution L_{fid}, the privacy risk L_{priv} and the cost κ low:

$$\theta^\star \in \underset{\theta:\, G_\theta \in \mathcal{F}(S,N)}{\arg\min} \big(-\Delta U,\ L_{\text{fid}},\ L_{\text{priv}},\ \kappa\big)(\theta). \qquad (1)$$

The constraint follows the data-quality dimensions of completeness and conformity [9], [10], and the objectives are the utility, fidelity and privacy on which synthetic data are evaluated [3], [11], together with cost.

Currently, deep generators [5], [12], [13] and language models fine-tuned on the target data [7], [14] must be trained on the scarce records, and they were designed and evaluated on datasets with tens of thousands of records. Approaches that prompt an LLM directly, such as CLLM [8] and EPIC [15], avoid training, but they focus on the design of the prompt and on the selection of the generated records; they do not ensure that the output ~~is a valid dataset of the requested size~~has the configured number of records and the correct columns and types, and they do not ~~measure~~ report its cost. ~~In addition~~Finally, generators are usually compared on the result of a single run, ~~although~~ which is unreliable in small data~~the variation between~~. A small test set amplifies chance: with 40 test records, one misclassified record changes accuracy by 2.5 percentage points, and the randomness of the data split and of the LLM's sampling makes two runs of the same generator ~~can be as large as the difference between two generators~~differ by amounts comparable to the differences being compared.

To address these limitations, we propose SYNTHAGENT ~~, a system that separates the generation of records, performed by the LLM, from their admission into the synthetic dataset, performed by~~ (*synthetic data agent*), an SDG system for small healthcare datasets that combines a pre-trained LLM, which proposes new records, with a *control layer*~~:~~ , a deterministic software component ~~placed between the LLM and the dataset, which~~ that checks every response of the LLM before any of its records is accepted into the synthetic dataset. The control layer selects the real records shown to the model, repairs and validates every response and asks for corrections, requests exactly the missing records, and logs the cost of every request. ~~The name (*synthetic data agent*) reflects this design: like~~ Like an LLM agent, the system thus alternates actions, the requests to the LLM, with observations, the checks of their output, which gives the system its name. This point is crucial since it makes the domain knowledge of LLMs usable where established generators are weakest, and since ~~a dataset with the requested size, valid columns and types, and a known cost~~ each healthcare facility can run the system on its own data and obtain a dataset that can be audited like any other data source.

The main contributions of this paper are:

- SYNTHAGENT, ~~a system for LLM-based SDG in small-data contexts~~ an SDG system for small healthcare datasets that requires no training on the target data ~~;~~
- ~~a control layerthat guarantees the requested~~ and, through its control layer, delivers the configured number of records ~~and their conformity to~~ with the columns and types of the real data ~~, and records the costof every request~~and a recorded cost;
- a deployment pattern in which each healthcare facility runs SYNTHAGENTlocally and only model updates are shared, so that no record leaves the facility;
- an open-source implementation that works with any OpenAI-compatible LLM endpoint, including locally served models.

~~Four~~ Our objective is to determine whether SYNTHAGENTmakes LLM-based SDG a reliable and useful option for small healthcare datasets. To this end, four main research questions emerged in this work:

RQ1 Is there any relationship between the control layer and the reliability of the generated datasets?

RQ2 Are the generated records faithful to the real data without reproducing them?

RQ3 Do the generated records improve the performance of classifiers trained on small data?

RQ4 How does SYNTHAGENT compare with established generation strategies, and how much do its results depend on the LLM?

To answer these questions, an experimental setup was built with six public healthcare datasets containing between 195 and 768 records. For each dataset, synthetic records were produced by SYNTHAGENT with four commercial LLMs, by direct prompting of the same LLM, with and without the provider's schema-constrained structured outputs, and by three established generation strategies, SMOTE, CTGAN and TVAE. Each dataset was randomly divided ten times into a training and a test set, and on each division ten classifiers were trained with and without synthetic records and compared on the same test set. The ~~answers are , in short,~~ main conclusions are the following. ~~RQ1: yes; direct prompting~~ Without the control layer, the LLM never returned the ~~requested~~ configured number of records, even when its decoding was constrained

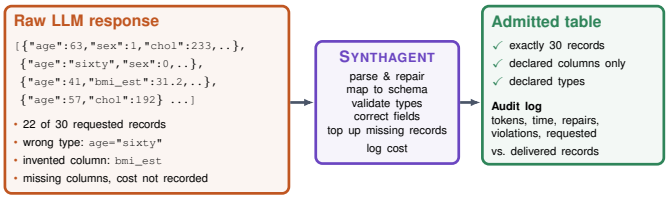


Fig. 1. The problem addressed in this paper, in an illustrative example. Asked for 30 records of a heart-disease dataset, an LLM response (left) can contain too few records, invented or missing features and values of the wrong type. SYNTHAGENT (right) validates and corrects every response, so the resulting dataset has the requested size and schema, and a log records each correction and its cost.

to the schema of the data, whereas SYNTHAGENT completed every dataset ~~to that number at a cost~~ at only 1.9% ~~higher. RQ2: the generated~~ more cost (RQ1). The synthetic records reproduced the distribution of each real feature with a fidelity of 73 out of 100, and a membership-inference attack could not reliably identify the real records used, although a classifier could still ~~distinguish~~ tell synthetic from real records ~~. RQ3: the effect~~ (RQ2). The effect on classifiers depended on the dataset and the classifier~~;~~, and with the strongest LLM ~~,~~ the average F1-score increased by 1.09 percentage points ~~. RQ4:~~ (RQ3). SYNTHAGENT gave the largest and most stable ~~F1~~ gain among the generators compared~~. Synthetic~~ : synthetic records can also ~~lower the performance of~~ harm a classifier when they misrepresent the real data~~; with the largest such loss on any dataset was 0.53 points, against~~ , and the established generators caused losses of up to 10.6 points ~~for the established generators. Only~~ on individual datasets, against at most 0.53 points for SYNTHAGENT; only the choice of LLM changed the results by more than the variation between repeated runs (RQ4).

The remainder of the paper is organized as follows. Section II presents the background knowledge ~~, Section ?? formalizes the problem,~~ and Section III describes SYNTHAGENT. Section IV presents the experimental setup, Section V reports the results, Section VI discusses their implications and limitations, and Section VII concludes the paper.

II. BACKGROUND KNOWLEDGE

In this section, the families of methods for synthetic data generation (SDG) of tabular data are first reviewed. Then, the limitations of each family when data are small are discussed, and finally the work on validating ~~the output of LLMs~~ generated data and on evaluating synthetic data, on which SYNTHAGENT builds, is presented.

A. Synthetic Data Generation for Tabular Data

Tabular SDG methods fall into four families, which differ in how they produce new records. *Interpolation and statistical methods* derive new records directly from the real ones or from a simple probabilistic model of them: SMOTE places a new minority-class record on the line segment between two neighbouring records of that class [4], and Bayesian networks estimate the dependencies between features and sample new records from them [16]. *Deep generative models* train a neural network to approximate the joint distribution of the features and then sample from it: CTGAN trains a generator against a discriminator, TVAE learns a compressed representation of the records [5], and diffusion models such as TabDDPM, CoDi and TabSyn learn to reverse a gradual corruption of the records [12], [13], [19]; relational and cross-silo variants extend these models to several tables or institutions [20], [21]. *Fine-tuned language models* write each record as a sentence, fine-tune a language model on these sentences and generate new ones, as in GReaT [7], REaLTabFormer [14] and TabLoft [22]. *Prompted LLMs* place a few real records in the prompt of a pre-trained LLM and ask it for new ones, without any training, as in CLLM [8] and EPIC [15].

B. Synthetic Data Generation in Small Data

Most of the previously mentioned generators were designed and benchmarked on datasets with tens or hundreds of thousands of records, such as Adult (48,842 records) or Covertype (581,012 records) [5], and their behaviour on a few hundred records has received far less attention. Table I summarizes, for each family, how records are produced and its main limitation when data are small. Interpolation and statistical methods, as well as deep generators, are fitted only to the real records, so they cannot add information beyond what those records contain [6]~~, and they ignore what the features mean~~. They also treat every column as an anonymous number: they do not know that a column holds a patient's age or cholesterol level, and therefore cannot use what is known about these variables. Deep generators and fine-tuned language models ~~must, in addition, estimate many parameters from few records~~have many parameters by design, which a few hundred records cannot estimate reliably, so they ~~can~~ may produce records outside the observed ranges and types, or reproduce training records [23]. Prompted LLMs need no training and bring domain knowledge, but nothing ensures that their output ~~is a valid dataset of the requested size, and its cost is not measured~~contains the configured number of records with the correct columns and types, and the cost of obtaining it is not reported; CLLM [8] filters the generated records by their learning dynamics and EPIC [15] designs balanced prompts, but both concern the quality of the records that are kept rather than the size, ~~validity~~ structure and cost of the output. ~~Across all families , finally, generators are~~ As discussed in Section I, all families are also usually compared on a single run~~, so differences caused by the variation between runs, which is large in small data, are reported as effects~~. SYNTHAGENT ~~targets the limitation~~ keeps the advantages of prompted LLMs ~~with its control layer while keeping their advantages~~and adds the control layer that they lack, and its evaluation ~~measures~~ repeats every experiment to measure the variation between ~~repeated runsbefore comparing configurations~~runs.

~~Families of tabular SDG methods: how they produce records, their main limitation in small data, and their properties. : by design; (): partially (SMOTE's number of records follows the class balance; fine-tuned language~~

TABLE I

FAMILIES OF TABULAR SDG METHODS: HOW THEY PRODUCE RECORDS, THEIR MAIN LIMITATION IN SMALL DATA, AND THEIR PROPERTIES. ✓: PROVIDED BY DESIGN; ×: NOT PROVIDED. SMOTE'S NUMBER OF RECORDS IS SET BY THE CLASS BALANCE, NOT BY THE USER; FINE-TUNED LANGUAGE MODELS DISCARD UNREADABLE RECORDS BUT DO NOT CHECK TYPES.

Family	How records are produced	Main limitation in small data	No training on target data	Exact size	Columns and types checked	Cost recorded
Interpolation and statistical [4], [16]	Combine real records or sample a simple model fitted to them	Only recombine existing patterns; ignore feature meaning	✓	×	×	×
Deep generative [5], [12], [13]	Train a neural network on the real records and sample from it	Few records to fit many parameters; invalid or copied records	×	✓	×	×
Fine-tuned language models [7], [14]	Fine-tune a language model on records written as text	Training on scarce records	×	✓	×	×
Prompted LLMs [8], [15]	Ask a pre-trained LLM for records, with real examples in the prompt	Unreliable output: wrong size, malformed records, unknown cost	✓	×	×	×
Prompted LLMs, constrained decoding [17], [18]	As above, with the output restricted to a JSON schema	Format guaranteed, but not the number of records; cost unknown	✓	×	✓	×
SYNTHAGENT (ours)	Prompted LLM whose output is checked and completed by a control layer	—	✓	✓	✓	✓

~~models discard unreadable records); ×: not addressed. Interpolation and statistical [4], [16] Combine real records or sample a simple model fitted to them Only recombine existing patterns; ignore feature meaning () × × Deep generative [5], [12], [13] Train a neural network on the real records and sample from it Few records to fit many parameters; invalid or copied records × × × Fine-tuned language models [7], [14] Fine-tune a language model on records written as text Training on scarce records × () × Prompted LLMs [8], [15] Ask a pre-trained LLM for records, with real examples in the prompt Unreliable output: wrong size, malformed records, unknown cost × × × (ours) Prompted LLM whose output is checked and completed by a control layer —~~

C. ~~*LLM Agents, Validated Output and* Validation of Generated Data~~*Quality*

The control layer ~~of , introduced in Section I, follows the agent pattern of alternating actions and observations [24], [25], but its observations are deterministic checks of the output of the LLM rather than the model's own reflections. This choice connects to a long line of data-management work. Synthetic data have long been used to test systems and share data [20], [26]; Kamino enforces denial~~ applies to generation a principle long used in data management: data produced by an automatic process are checked against explicit constraints before they are accepted. Kamino enforces integrity constraints during private data synthesis [27]~~; ,~~ data-cleaning systems combine statistical inference with integrity constraints [28], [29]~~; foundation models can clean and integrate data through prompting [30] ; ,~~ and recovering structured records from the free-form ~~LLM output is~~ output of an LLM has been studied as a data-engineering problem~~in its own right~~ [31]. Constrained decoding goes further and restricts the tokens an LLM may emit to those allowed by a grammar or a JSON schema [17], [18], and LLM providers now offer it as a structured-output mode. It guarantees the format of each response, but not how many records a response contains, so it does not by itself deliver a dataset of a given size; Section V-A tests it as a baseline. In SYNTHAGENT ~~applies the same principle to generation: the schemaand type rules are~~ , the constraints are the columns and types inferred from the real data, and ~~every response is parsed and validated against them. Unlike feature augmentation, which adds columns from related tables [32], it adds records~~the checks are applied to every response of the LLM, in the action–observation loop of LLM agents [24], [25].

D. Evaluating Synthetic Data

Data-preparation studies have shown that a step that looks beneficial can harm downstream models [33], [34], and evaluation frameworks such as Synthcity and SynthEval recommend assessing fidelity, privacy and utility together [3], [11]. Fidelity can be complemented by a classifier two-sample test, which asks whether a classifier can tell real from synthetic records [35], and privacy by membership-inference attacks, which ask whether an attacker can tell which records were used for generation [23], [36]. Formal guarantees require syntactic anonymization models such as ℓ-diversity and t-closeness [37], [38] or differential privacy [39], which SYNTHAGENT does not claim. ~~Our protocol (Section IV) adopts the multi-dimensional view and adds the paired comparison and noise estimate that small data requires.~~

III. Problem Formulation

We now state the problem introduced in Section I precisely. Given a small dataset D, its schema S (the features and their types) and a requested number of synthetic records N, the goal is to produce, without training a generator on D, a synthetic dataset G that satisfies six requirements: it has exactly N records (R1); every record conforms to S (R2); the cost of producing it is measured (R3); it is faithful to the distribution of D (R4); it does not reproduce the records of D (R5); and adding it to the training data improves, or at least does not degrade, the classifiers trained on D (R6). These requirements are not new; they combine two established lines of work. R1 to R3 correspond to data-quality dimensions used in data management, namely completeness, conformity to a schema and the cost of acquiring the data [9], [10], and they concern the synthetic dataset as a data product: a system can guarantee them. R4 to R6 are the three dimensions on which synthetic data are evaluated in the literature, namely fidelity, privacy and utility [3], [11], and they concern the value of the synthetic records: they can only be measured. The relation between the two groups is therefore one of order. A dataset that fails R1 or R2 cannot be used at all, so its fidelity, privacy and utility are not even defined, whereas a dataset that satisfies R1 to R3 can then be judged on R4 to R6. Formally, let D^{tr} and D^{te} be a training and a test partition of a small dataset D with schema S, and let a configuration θ of the system (LLM, context strategy, active rules) produce a synthetic dataset G_θ from D^{tr} without access to D^{te}. Requirements R1 and R2 define the feasible set $\mathcal{F}(S, R, N)$ of datasets with exactly N records that conform to S and to the active rules R. The other requirements are objectives: a fidelity loss $L_{\mathrm{fid}}(D^{\mathrm{tr}}, G_\theta)$ for R4, a privacy risk $L_{\mathrm{priv}}(D^{\mathrm{tr}}, G_\theta)$ for R5, the change in downstream utility $\Delta U(G_\theta)$ for R6 and the generation cost $\kappa(\theta)$, which R3 requires to be measured and which should also be kept low: The objectives in Equation (1) can conflict: records closer to the real ones are more faithful but easier to attack, and a larger LLM may produce better records at a higher cost. We therefore do not collapse them into one weighted score, because the right weights depend on the use; a healthcare facility sharing data for software testing weights privacy and validity, whereas a team augmenting a classifier weights utility. The formulation also divides the work of the paper. The constraint $G_\theta \in \mathcal{F}(S, R, N)$ is the responsibility of the system, which the next section describes; the objectives are the subject of the evaluation that follows it.

III. The SynthAgent System

SynthAgent meets the constraint of Equation (1) by checking the output of the LLM instead of relying on it: every response is verified, and corrected or completed when necessary, before its records enter the synthetic dataset; the real dataset is never modified. The architecture that implements this principle is shown in Figure 2. A profiling step infers the schema S and the type rules R columns and types S from the training datarecords. Each request then passes through four stages: the context selector builds the prompt (1), the LLM proposes a batch of records (2), the parser and validator check it (3), and the accepted records are counted against the requested size configured number N (4). Each type of failure triggers its own correction. We describe first the state that ties these stagestogether and then each stage in turnThe fixed parameters of these stages, and the reason for each value, are listed in Table II; they were set before the experiments and not tuned.

TABLE II
Fixed parameters of the control layer and the reason for each value. All were set before the experiments and not tuned; the LLM-chosen settings variant (Section IV) lets the LLM choose the first and fourth instead.

Parameter	Value	Reason
Context sample ρ	20% of training records	31 to 123 records in our datasets: enough to show the main patterns of the data, while the prompt still fits the context window of every LLM tested
Window share ω	85%	Leaves room for the answer and for errors in the estimated prompt length
Batch size b	30 records	The longest answer (30 records of the 30-feature dataset) stays within the output limit of every LLM tested; a failed answer loses at most 30 records
Temperature	0.3	Favours adherence to the output format while keeping some diversity
Repair temperature	0.1	A repair should fix the syntax, not change the content
Correction rounds c	2	Bounds the cost of a batch that keeps failing
Top-up rounds $t_{\max}$	50	$50 \times 30 = 1{,}500$ records, more than twice the largest N (614)
Empty-answer limit $u_{\max}$	5	Five consecutive empty answers indicate a provider failure rather than a sampling accident

A. State of the Control Layer

Let N be the requested number of synthetic records and R the active rules. At step t, the control layer keeps the state

$$z_t = (S,\, C_t,\, A_t,\, N - |A_t|,\, B_t,\, E_t), \tag{2}$$

where C_t is the set of real records shown to the LLM, A_t the accepted synthetic records, $N - |A_t|$ the number still missing, B_t the remaining token budget and E_t the log of parser and validator results. The LLM, denoted π_ϕ, proposes a batch $Y_t \sim \pi_\phi(\cdot \mid z_t)$. Let P_S be the parser, which also maps the returned features columns to S, $V_{S,R}$ V_S the validator and K a correction step that sends invalid values back to the LLM, at most twice. The accepted set is updated as

$$A_{t+1} = A_t \cup K_{V_S}\big(P_S(Y_t)\big). \tag{3}$$

Equation (3) expresses the central design decision: the LLM only proposes, and a record enters A_{t+1} only after it has been parsed, mapped to S and validated. Because the state is explicit rather than implied by the conversation history, the number of missing records is always known, and E_t records every failure for later inspection. The remaining subsections describe how each component reads and updates this state.

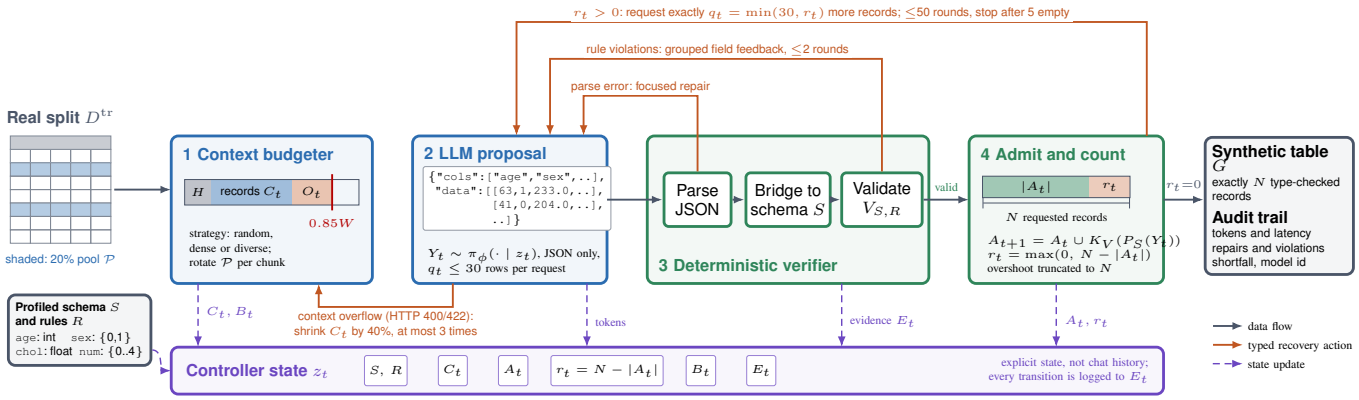


Fig. 2. Architecture of SYNTHAGENT. Real records enter a context window of limited size (1); the LLM proposes a JSON batch (2); deterministic checks parse it, map it to the schema and validate the type rules (3); and accepted records are counted against the requested size (4). Each failure type triggers its own correction (orange). Every stage reads or updates the state z_t of the control layer (purple), which also serves as the audit log of the run.

B. Context Selection~~(R1, R3)~~

The context selector decides which real records the LLM sees. Table III lists exactly what the model receives~~in our experiments~~. The core evidence is a ~~20% sample of~~ sample of $\rho = 20\%$ of the real training ~~split, serialized as compact row-major~~ records, between 31 and 123 records in our datasets, written as compact JSON with the ~~ordered~~ feature columns and the ~~observed target, so~~ class, so that the model sees how features and class co-occur~~in real records~~. No free-text description of the prediction task is added, so that no dataset receives more prompt engineering than another~~; the model must infer the role of each feature from its name and values~~.

~~Formally, a selection strategy $s \in \{\text{random, dense, diverse}\}$ defines an ordered candidate pool~~A selection strategy orders the sample, and the ~~token budget determines how much of that pool is visible:~~ selector shows the model as many records as fit in a share ω of the LLM's context window W:

$$C_t = \text{Slice}_t\big(\text{Select}_s(D, \lceil \rho|D| \rceil), k_t\big),$$
$$k_t = \max\{k : T(C_{t,k}) \le \omega W\}, \tag{4}$$

~~Here H is the schema and output contract, I optional instruction text, O_t the reserved output, W the backend context window and~~ where T ~~a token estimator (2.8 characters per token for compact JSON)~~estimates the length in tokens of a prompt that contains k records. The slice ~~operator~~ rotates through the ~~pool across chunks~~sample across batches, so successive requests see different patients~~when the pool exceeds one window~~. If the provider still rejects a request as too long~~(HTTP 400 or 422), the control layer shrinks the visible context by 40% and retries, at most three times. "20% context" therefore describes the candidate pool, not a promise that every candidate row appears in every prompt.~~

~~The 20% fraction and the 85% window limit were fixed before the experiments and not tuned~~, the selector shows fewer records and retries. The default strategy samples ~~the context at random.~~ at random; ~~*Dense*~~*dense* prefers records with ~~the most observed cells, reducing ambiguity from missing fields.~~ ~~*Diverse* standardizes numerical features, runs k-means and~~ fewer missing values, and *diverse* selects the records ~~nearest to the centroids~~closest to the centres of k-means clusters, so that the ~~context~~ sample covers more of the feature space. Neither strategy is class-stratified, ~~so neither guarantees that a rare diagnosis appears in context;~~ a limitation discussed in Section VI~~returns to this limitation~~.

C. Parsing, Validation and Correction~~(R2)~~

~~The parser and the validator decide which parts of a response can enter the accepted set A_t of Equation (3).~~ Records are requested in ~~JSON batches of at most 30, a size that keeps every response within the output limit of the smallest LLM tested and limits the loss when a response fails~~batches of $b = 30$. A local parser ~~first extracts structured content~~ extracts the records from each response and fixes common ~~defects (non-finite literals, trailing commas, concatenated objects). If~~ syntax defects; if parsing still fails, the control layer ~~issues~~ sends one repair request ~~at temperature 0.1~~ that quotes the ~~error position and surrounding text~~position of the error. The returned ~~feature~~ column names are then mapped back to ~~the schema.~~

~~Profiling infers a type rule for every feature, and optional boundary rules (numerical ranges with 10% tolerance, or categorical domains with at most 20 values). Dynamic Pydantic models validate each batch. The validator groups failures by field and reports one representative bad value per field, so the correction prompt does not grow linearly with repeated errors. Up to two correction roundsare attempted per batch. User-defined post-processing rules (noise, rounding, normalization) can be applied deterministically afterwards.~~

~~In the reference configuration of our experiments, type rules are active, boundary rules are off and no user rules are supplied, so we evaluate the default system rather than the strictest one. Rules are used only by the validator and are never inserted into the prompt~~S. The validator checks that every value has the type inferred for its column; optional rules can also bound numerical ranges or restrict categories, but they were off in our experiments. Failures are grouped by column

TABLE III
INFORMATION BOUNDARY FOR ONE GENERATION REQUEST IN THE REPORTED STUDY. "NOT SUPPLIED" IS INTENTIONAL: THOSE SIGNALS CANNOT INFLUENCE THE GENERATED RECORDS.

Context layer	What the model receives	Why it is included or excluded
Schema	Ordered column names and an exact column-count requirement	Fixes the record interface and prevents invented or omitted fields.
Demonstrations	Compact row-major serialization of a 20% sample of the training records, with feature and target values	Exposes joint patterns, value formats, label prevalence and plausible cross-column combinations without sending the whole local table.
Generation task	Exact number of new records for the current batch, at most 30	Turns an open-ended completion into a bounded action whose output count can be verified.
Output contract	JSON only, fixed column order, complete cells, raw numeric values, no extra metadata	Makes parsing deterministic and removes interface failures unrelated to statistical quality.
Dynamic feedback	Parser error location after malformed JSON, or grouped field violations after failed validation	Reveals only the evidence needed for a targeted repair instead of replaying the full history.
Free-text metadata	Not supplied	Prevents unequal, dataset-specific prompt engineering from confounding the comparison.
Test information	No holdout rows, predictions, metrics or classifier identities	Preserves the evaluation boundary and prevents optimization against the test set.
Global statistics	No separately computed means, quantiles, correlations or class quotas	Fidelity must arise from the selected records; the consequence is weaker protection against rare-mode omission.

and sent back to the LLM with one example of an invalid value per column, for at most $c = 2$ correction rounds.

D. Completion Loop ~~(R1)~~

LLMs often return fewer records than requested~~, as RQ1 will show. The completion loop derives its next request from the accepted state rather than from the conversation: Equation (5)~~. After the planned batches, the completion loop requests only the ~~residual q_t, and any overshoot is truncated. The loop runs for at most 50 top-up rounds and stops after five consecutive empty or unparsable responses. Successful runs therefore return exactly~~ missing records,

$$r_t = \max(0, N - |A_t|), \qquad q_t = \min(b, r_t), \tag{5}$$

and truncates any excess. It stops when N ~~records; persistent provider failure can still end a run short, and that shortfall is recorded rather than hidden~~records have been accepted, after $t_{\max}$ rounds, or after $u_{\max}$ consecutive empty answers; a run that ends short because the provider keeps failing is recorded as such. Algorithm 1 summarizes the complete ~~loop.~~

~~Generation loop of the control layer~~

~~The last element of the control layer is its audit log, which serves R3~~procedure, with the parameters of Table II. Every run ~~records prompt, completion and total tokens , LLM~~ also records the tokens processed, the generation time, ~~parser repair attempts, rule violations ,~~ the repairs, the violations and the requested and delivered ~~record counts and the model identifier. These~~ numbers of records; these records make a synthetic dataset inspectable after the fact and give the cost ~~term $\kappa(\theta)$~~ κ in Equation (1) a measured value.

E. Deployment Across Healthcare Facilities

~~The setting that motivates this work, small datasets held by healthcare facilities that cannot pool them~~As discussed in Section I, healthcare facilities often cannot pool their records [1], [40], ~~also suggests how~~ and SYNTHAGENT ~~can be deployed~~was designed for this setting. Because the control layer needs only a local dataset, its schema and an inference endpoint, each healthcare facility of a network can run it inside its own infrastructure. Consider K healthcare facilities, each holding a small dataset D_k for the same prediction task, with different case mix, coding practice and class prevalence. Each healthcare facility runs SYNTHAGENT locally and trains on its real and synthetic records,

$$G_k = \mathcal{A}(D_k, S_k, N_k), \qquad \Delta w_k = \mathcal{U}(w_t, D_k \cup G_k), \tag{6}$$

where S_k ~~and R_k~~ are the local ~~schema and rules, B_k the local token budget~~ columns and types, N_k the local number of synthetic records and $\mathcal{U}$ the local training procedure. Only the model update Δw_k and approved aggregate diagnostics leave the healthcare facility. A coordinator aggregates the updates as in FedAvg [41], [42],

$$w_{t+1} = w_t + \sum_{k \in \mathcal{K}_t} \frac{n_k}{\sum_{j \in \mathcal{K}_t} n_j} \Delta w_k, \qquad n_k = |D_k|, \tag{7}$$

preferably under secure aggregation [43]. The weights count *real* records, $|D_k|$ rather than $|D_k \cup G_k|$, so a healthcare facility gains no extra weight by generating more records (Figure 3). Unlike SiloFuse, which trains a shared generator across sites [21], this design needs no generator training across

Algorithm 1 Generation procedure of SYNTHAGENT

Require: training records D, columns and types S, number of records N, parameters of Table II

$\mathcal{P} \leftarrow$ sample of $\rho|D|$ records; $A \leftarrow \emptyset$
repeat
  $C \leftarrow$ next records of $\mathcal{P}$ that fit in ωW ▷ Eq. (4)
  $Y \leftarrow$ ask the LLM for $\min(b, N - |A|)$ records, given S and C
  $X \leftarrow$ parse Y and map it to S, repairing it if needed
  correct the invalid values of X, for at most c rounds
  $A \leftarrow A \cup X$; log tokens, repairs and violations
until $|A| \geq N$ **or** $t_{\max}$ rounds **or** $u_{\max}$ empty answers
return the first N records of A and the log

TABLE IV

EVALUATION DATASETS. n: RECORDS; d: FEATURES; k: CLASSES; $n_{\min}$: RECORDS IN THE SMALLEST CLASS; $n^\star$: MINIMUM SAMPLE SIZE BY THE FIRST CRITERION OF RILEY ET AL. [2] (SHRINKAGE 0.9, $d(k-1)$ PARAMETERS, ANTICIPATED R^2_{CS} OF 15% OF ITS MAXIMUM); RATIO: $n^\star/n_{\mathrm{tr}}$, WITH $n_{\mathrm{tr}} = 0.8n$ THE TRAINING SIZE.

Dataset	n	d	k	$n_{\min}$ (%)	$n^\star$	Ratio
Heart disease	303	13	5	13 (4.3)	10,201	42.1
Breast cancer	569	30	2	212 (37.3)	2,302	5.1
Heart failure	299	12	2	96 (32.1)	946	4.0
Diabetes	768	8	2	268 (34.9)	621	1.0
Liver disease	583	10	2	167 (28.6)	808	1.7
Parkinson's	195	22	2	48 (24.6)	1,851	11.9

For heart disease, $n^\star$ applies the binary criterion to the smallest class versus the rest and is an approximation. Parkinson's contains several recordings per subject, so its effective number of independent records is lower and its ratio higher than shown.

healthcare facilities, and each healthcare facility can reject its synthetic records when paired local validation shows no gain. The design is not, by itself, a privacy guarantee: with a remote LLM the prompt content leaves the healthcare facility, so sensitive deployments need a locally served model or a data-processing agreement, and both synthetic records and model updates can leak information [44]. Our experiments evaluate the generation step one healthcare facility at a time, which is the setting the next section describes; a ~~multi-site~~ multi-facility evaluation is future work.

IV. EXPERIMENTAL SETUP

~~The system guarantees R1–R3 by construction; R4–R6 must be measured, and in small data measuring them reliably is itself a challenge.~~ This section describes the data, the protocol, the configurations and the measures with which ~~we answer RQ1–RQ4, and explains why each choice was made~~RQ1 to RQ4 are answered.

A. Datasets

~~We use healthcare data because small data is most common there and because the case for SDG, more training records without exposing patients, is strongest [1]. We use public datasets so that every result can be reproduced.~~ Six public classification datasets were selected (Table IV~~lists the six classification datasets~~): five from the UCI repository [45]–[49] and the Pima Indians diabetes benchmark [50]. They are widely used, and each has fewer training records than the minimum sample size recommended for its task by the criteria of Riley et al. [2]~~(ratio $n^\star/n_{\mathrm{tr}} \geq 1$), and together~~. Together they span the factors that make small data harder: 195 to 768 records, 8 to 30 features, two to five classes ~~,~~ and a smallest class from 4.3% to 37.3% of the records~~, and ratios between 1.0 and 42.1~~. We keep the original five-class heart-disease target and impute the few missing values (0.1% of the cells in two datasets) with the mean for numerical and the mode for categorical features.

B. Protocol

Measuring whether synthetic records help a classifier requires real records that the generator has never seen~~, so each dataset is split~~. Each dataset is therefore divided into a training ~~partition~~set, given to the generator, and a test ~~partition~~set of 20% of the records, used only to score classifiers. ~~A single split, however, is unreliable in small data: the test partition holds only~~ Because a test set of 39 to 154 records ~~, and the result changes with the split. We therefore repeat the split~~ makes a single division unreliable, the division is repeated ten times with ~~seeds 43 to 52 (80/20 holdouts) and report~~ different random seeds, and means and standard deviations are reported. In each repetition~~the generator receives the training partition and produces as many synthetic records as there are training records; the LLM sees a 20% sample of the training partition as context and generates at temperature 0.3, a value fixed in advance to favour adherence to the output format while keeping some diversity.~~

~~The utility of synthetic data also depends on the learner: interpolated records can help a nearest-neighbour classifier and harm a boosted ensemble. To avoid conclusions tied to one learner, we train ten classifiers that cover~~ , the number of synthetic records N equals the number of training records, so augmentation doubles the training set. Ten classifiers covering the main model families ~~: linear (logistic regression~~ , are trained: logistic regression (LR), ~~kernel (~~support vector machine ~~,~~ (SVM), ~~distance-based (~~k-nearest neighbours ~~,~~ (kNN), ~~probabilistic (~~Gaussian naive Bayes ~~,~~ (NB), ~~a single tree (decision tree ,~~ decision tree (DT), ~~bagging (random forest ,~~ random forest (RF), ~~boosting (gradient boosting , GB; AdaBoost; XGBoost) and a neural network (multilayer perceptron ,~~ gradient boosting (GB), AdaBoost, XGBoost and a multilayer perceptron (MLP). Using several families avoids conclusions tied to one learner, since interpolated records can help a nearest-neighbour classifier and harm a boosted ensemble. Each classifier is trained ~~in three regimes and scored on the same test partition:~~ on real records only, on synthetic records only, and on real plus synthetic records, and all three are scored on the same test set. XGBoost failed on liver disease in every repetition because the labels are coded $\{1, 2\}$~~; the code now re-encodes them, but filling this cell would require regenerating every configuration, so we report~~ , so 59 of the 60 classifier–dataset ~~cells~~combinations are reported.

C. Configurations

~~We evaluate 14 configurationsand analyse only those that~~ Table V lists the 17 configurations. A configuration is included in the analysis only if it completed all six datasets and ten repetitions. The ~~first group answers RQ1. A~~ *reference* configuration is SYNTHAGENTwith its default settings and GPT-4o mini, the least expensive LLM; every other configuration differs from it in one respect. The *direct-prompting* baseline uses the same LLM, prompts and context~~as ,~~ but none of ~~its control layer: the response is parsed with a strict JSON parser, and there is no repair, schema mapping, correction, retry or completion loop. Seven *control-layer variants* then change one element of the pipeline at a time: the reference ; three variants that disable a component (type rules, post-processing, and a legacy context-join option); dense and diverse context selection; and~~

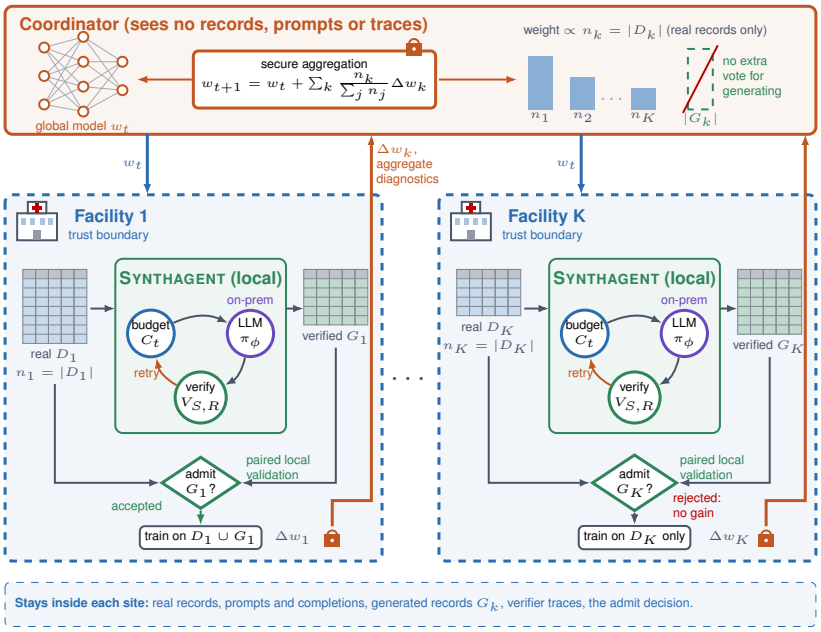


Fig. 3. Deployment across healthcare facilities. Each healthcare facility runs SYNTHAGENT (context selection, a locally served LLM, validation and correction) inside its own infrastructure, then decides with paired local validation whether to use the synthetic records G_k (Facility 1) or to train on real data only (Facility K). Only the model update Δw_k and approved aggregate diagnostics leave the healthcare facility. The coordinator weights each update by the number of real records $n_k = |D_k|$ (Equation (7)), so generating more records gives a healthcare facility no extra weight.

~~a variant in which~~ the control layer, and the ~~LLM chooses context size, selection strategy and temperature from a profile of the dataset. All eight use GPT-4o mini, the least expensive LLM, so that every comparison runs at the lowest cost.~~

~~The second group answers RQ4. Four *LLMs* run the reference configuration:~~ *structured-output* baseline adds to it the provider's strict JSON-schema mode, which constrains decoding to the column names, numerical types and observed categories of the real data. Eight control-layer variants change one component or repeat the reference run, and three LLM variants replace GPT-4o mini, ~~GPT-4.1 mini, GPT-4o and GPT-4.1,~~ forming a 2×2 design that crosses model generation (4o versus 4.1) with model size (mini versus full)~~. Drawing them~~ ; all four LLMs come from one provider~~keeps the interface, tokenizer and output mode constant~~, so that ~~differences can be attributed to the model; the control layer itself works with any OpenAI-compatible endpoint, including locally served open models. Three *established generators* use the same splits: SMOTE [4], which balances every class to the majority count and therefore emits only non-majority records,~~ the interface and tokenizer are constant. Three established generators, SMOTE and CTGAN and TVAE ~~[5]~~ with the default settings of ~~the reference implementation (300 epochs)~~their reference implementations, use the same divisions. In total, the ~~14 configurations give 840~~ configurations give 1,020 dataset–configuration runs and up to ~~25,200~~ 30,600 classifier fits.

Three of the control-layer variants ~~turn out to~~ change nothing in this setting~~. With~~ : with no user rules, disabling post-processing removes a step that does nothing~~;~~ , the context-join option affects only a legacy fidelity score~~;~~ , and disabling type rules cannot change the output because the validator never found a type violation with GPT-4o mini. Together with the reference ~~, these variants are therefore four~~ and a rerun of it that only adds per-request logging, they are five independent runs of the same pipeline, and ~~we use them as such:~~ their spread estimates the variation between repeated runs, which we call the *noise floor*~~. A~~ : a difference between two configurations that is smaller than this spread is not evidence of an effect.

TABLE V
THE 17 EVALUATED CONFIGURATIONS. ALL LLM CONFIGURATIONS EXCEPT THE LLM VARIANTS USE GPT-4O MINI.

Group	Configuration	Purpose
Baselines (RQ1)	Direct prompting; structured outputs	Same LLM without the control layer, with free or schema-constrained decoding
Reference	SYNTHAGENT, default settings	Point of comparison for all others
Repeated runs (RQ1)	No type rules; no post-processing; no context join; logged rerun	Change nothing in this setting; estimate the variation between runs
Design choices (RQ1)	Dense context; diverse context; LLM-chosen settings; range enforcement	Effect of context selection, of letting the LLM set ρ and the temperature, and of clipping values to the observed ranges
LLM variants (RQ4)	GPT-4.1 mini; GPT-4o; GPT-4.1	Effect of model generation and size
Established generators (RQ4)	SMOTE; CTGAN; TVAE	Generators fitted to the real records

D. Measures

~~Each requirement of the problem statement has its own measure. For *reliability*~~*Reliability* ~~(R1–R3), we record~~ is measured by the number of records delivered against the number ~~requested~~configured, the number of malformed responses, ~~completeness (~~the percentage of non-empty values~~)~~, the corrections made ~~by the control layer~~ and the cost in

tokensand generation time. For , *fidelityFidelity* (R4), we compare compares the synthetic records alone , without the real context, with the real training records over the numerical features $\mathcal{J}$:

$$F_{\mathrm{JSD}} = 100\Big(1 - \tfrac{1}{|\mathcal{J}|}\textstyle\sum_{j\in\mathcal{J}} D_{\mathrm{JS}}(D_j, G_j)\Big),$$
$$F_{\mathrm{corr}} = 100\big(1 - \mathrm{mean}\,|\Sigma_D - \Sigma_G|\big), \tag{8}$$

where D_{JS} is the Jensen–Shannon divergence between real and synthetic marginals distributions of a feature and Σ_D, Σ_G are their the correlation matrices; both scores equal 100 for identical distributions. We add a A classifier two-sample test (C2ST) [35] , in which trains a random forest is trained to separate real from synthetic recordsand ; an accuracy close to 0.5 means that they are hard to tell apart, and, for . For the established generators and direct prompting , we also report the percentage of *valid* records, whose values lie in the observed ranges and categories and are integers where the real feature is, a stricter criterion than R2, which concerns features and types only. For the column-and-type checks of the control layer. *privacyPrivacy* (R5), is measured by a membership-inference attack that scores each real record by its distance to the nearest synthetic record and reports the ROC AUC between training and test records, where 0.5 is chance. For *utilityUtility* (R6), is the paired change, for metric $m \in \{\mathrm{accuracy}, \mathrm{F1}\}$, dataset d, classifier c and repetition $h$is ,

$$\Delta m_{d,c,h} = m\big(D_{d,h} \cup G_{d,h};\, c\big) - m\big(D_{d,h};\, c\big), \tag{9}$$

with both models scored on the same test partition, so that variation between test partitions is not mistaken for an effect of SDGset. We use binary F1 for the binary tasks and class-weighted F1 for heart disease; binary F1 scores the class coded 1, which is the minority class for heart failure and diabetes and the majority class for liver disease and Parkinson's. Changes are reported , and report changes in percentage points (pp).

E. Statistical Analysis

Repetitions and classifiers are not independent datasets, so we average them they are averaged within each dataset and treat the six datasets are treated as matched blocks. Configurations are compared with the Friedman test,

$$\chi_F^2 = \frac{12M}{Q(Q+1)} \sum_{a=1}^{Q} \bar{\mathrm{rk}}_a^{\,2} - 3M(Q+1), \tag{10}$$

where $\bar{\mathrm{rk}}_a$ is the average rank of configuration a over $M = 6$ datasets and Q the number of configurations, followed by two-sided Wilcoxon signed-rank tests of the best-ranked configuration against every other, with Holm correction [51], [52]. Per-cell tests of augmented versus real-only training over the ten repetitions are Holm-corrected within each metric. With six blocks the tests have limited power, which is precisely why we report so effect sizes and the noise floor are reported alongside p-values. With these measures in place, we can now answer the research questions in order.

V. Results

A. RQ1: What the Control Layer Delversand Reliability

The first question is whether the control layer is needed at all, and the direct-prompting baseline answers itcontrol layer and the reliability of the generated datasets are directly related: without it, the same LLM never delivered the configured number of records; with it, every dataset was complete, at 1.9% more tokens and with unchanged quality. Table VI compares it direct prompting with the reference SYNTHAGENT , both with GPT-4o mini, on the same prompts, context and splitsdivisions. Direct prompting returned the requested number of records in none of the 60 runs. It delivered on average 87.6% of the configured records, fell short in 50 runs, by 4% to 43%, of 60 runs and overshot in the other 10diabetes runs, by up to 3.4%. On average it delivered 87.6% of the requested records, and as little as 64.0% on the 30-feature breast-cancer dataset, where the model often returned fewer than the 30 records requested per batch. Ten responses, eight on breast cancer and two on Parkinson's, were not valid JSON and , and lost an entire batch eachto each of ten malformed responses. SYNTHAGENT delivers exactly the requested number by construction, N records because its completion loop requests the missing records ones and truncates any excess, and its parser recovers malformed responses instead of discarding them.

Direct prompting versus (RQ1), both with GPT-4o mini, over ten splits. Delivered: records returned as a percentage of those requested (range over splits); Malf.: malformed responses; ΔF1: mean F1 change (pp).

The control layer achieves this without changing the quality of the records. The two configurations have the same marginal fidelity Fidelity, privacy and utility were the same for both configurations (JSD 72.7), similar correlation preservation (71.2 versus 72.6) and C2ST accuracy (0.883 versus 0.878), a membership AUC near chance (; attack AUC 0.517 versus 0.526) and mean utility changes ; F1 change -0.04 versus -0.14 pp, within the noise floor estimated below(-1.36 versus -1.19 pp in accuracy, -0.04 versus -0.14 pp in F1). The direct responses contained no type errors or unknown features with this LLM , so its failures were failures of volume and syntax, exactly the ones the completion loop and below), so the difference lies not in the records but in whether the dataset that was configured is delivered.

A natural objection is that the structured outputs now offered by LLM providers, which constrain decoding to a JSON schema, make such a layer unnecessary. They do not replace it. With the same prompts and a strict schema fixing the column names, the parser address. The cost of the control layer is small: 73numerical types and the observed categories, every response parsed and no value had the wrong type, but the LLM returned on average only 77.7% of the configured records (51–97% over divisions), never the full dataset; 11 requests failed outright, and the run used 21% more tokens than SYNTHAGENT. The schema also weakened the dependence between features: correlation fidelity fell from

TABLE VI
DIRECT PROMPTING, STRUCTURED OUTPUTS AND SYNTHAGENT(RQ1), ALL WITH GPT-4O MINI, OVER TEN DIVISIONS. DELIV.: RECORDS RETURNED AS A PERCENTAGE OF THE CONFIGURED NUMBER N (RANGE OVER DIVISIONS FOR DIRECT PROMPTING); MALF.: MALFORMED RESPONSES; ΔF1: MEAN F1 CHANGE (PP). SYNTHAGENTDELIVERED 100% IN EVERY RUN.

		Direct prompting			Structured		SYNTHAGENT
Dataset	N	Deliv. (%)	Malf.	ΔF1	Deliv.	ΔF1	ΔF1
Heart disease	242	93.5 (91.7–94.6)	0	−0.15	82.1	−0.88	−0.97
Breast cancer	455	64.0 (56.9–69.0)	8	−0.78	63.9	−0.97	−0.62
Heart failure	239	94.5 (93.3–95.8)	0	+0.26	79.3	+2.65	−1.00
Diabetes	614	102.2 (101.3–103.4)	0	+2.23	93.2	+2.90	+2.39
Liver disease	466	93.8 (92.7–95.5)	0	+0.12	76.8	−0.71	+0.13
Parkinson's	156	77.3 (62.2–87.8)	2	−1.94	70.8	−0.73	−0.77
Mean		87.6	10	−0.04	77.7	+0.38	−0.14

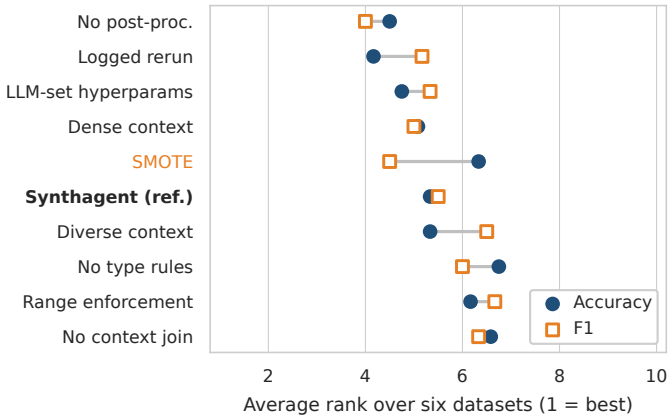


Fig. 4. Average ranks of the control-layer variants and SMOTE over the six datasets (lower is better). Neither accuracy nor F1 shows a significant difference.

72.6 to 63.0 and the C2ST accuracy rose from 0.878 to 0.943, so the records became easier to tell from real ones, while the F1 change (+0.38 pp) stayed within the noise floor, and a Friedman test over the three configurations finds no difference in utility ($p = .846$ for accuracy, $p = .311$ for F1). Constrained decoding therefore guarantees the format of each record, but not the size of the dataset, 510 tokens per repetition against 72, 126, or 1.9% more . The two configurations differ, in other words, not in what the records look like but in whether the user receives the datasetthat was requested. which is what the completion loop of the control layer provides.

Having established that the control layer is needed, we ask which of its design choices change the outcome.

The design choices inside the control layer matter much less than the control layer itself (Table VIIcompares the seven control-layer variants. The first four rows run). The five repeated runs of the same pipeline , so their spread is the noise floor: JSD 72.43–72.92, correlation 70.83–72.93, mean accuracy change −1.80 to −1.19 pp, mean F1 change −0.18 to +0.28 pp and attack AUC 0.523–0.534. Differences of about differ by up to 0.5 JSD points, 2 correlation points or 0.6 and 0.7 pp of utilitybetween two runs are therefore not evidence of a design effect. The same rows show that generation time is a poor cost measure: the four equivalent runs take 228 to 379 seconds per repetition, depending on the provider, while their token counts differ by less than 2%. We therefore measure cost in tokens.

Control-layer variants (RQ1), all with GPT-4o mini. The first four rows run the same pipeline and define the noise floor. Seconds and tokens are means per repetition. Bold: best value among the LLM configurations; lower is better for C2ST, for the distance of the attack AUC from 0.5, and for cost.

Against this floor, the design choices have small effects, which is the noise floor. Dense context stays inside the floor on every measure. Diverse context is the only choice that leaves iton fidelity , with the highest JSD (73.27) and correlation it; diverse context slightly improves fidelity (JSD 73.27, correlation 73.72) and the lowest C2ST accuracy (0.872), but also the highest at the cost of a slightly higher attack AUC (0.544): records that cover more of the feature space lie closer to more real records. Letting ; letting the LLM choose its own settings stays inside the floor on fidelity and utility and changes nothing but costs 57% more tokens(115, 245 versus 73, 510); and clipping values to the observed ranges lowers JSD by 0.9 points, just beyond the noise floor, without improving utility. The Friedman tests find no difference in utility between the variants ($\chi^2_F(7) = 3.10 p = .843$ for accuracy, $p = .875$ for accuracy; $\chi^2_F(7) = 4.39$, $p = .734$ $p = .872$ for F1), and every Holm-adjusted comparison has $p = 1.0$. The ranks in ; Figure 4show how little the variants separate: LLM-chosen settings rank best on accuracy (3.75), and the no-post-processing variant, which changes nothing, ranks best on F1 (3.33). SMOTE ranks sixth of eight on accuracy (5.00) and second on F1 (3.67). Generation time varied by a factor of 1.7 between the repeated runs, depending on the provider, while their token counts differed by less than 2%, so cost is measured in tokens.

The correction steps were rarely needed with corrections are what allow a less disciplined LLM to be used. GPT-4o mini : it produced no type violation, and the local parser fixed every defect in its responseswithout a repair request. They were needed with . GPT-4.1 mini, the fastest LLMwe tested, which , produced on average 7.4 type violations per repetition on breast cancer and one malformed responsein ten repetitions on diabetes. These were detected and ; these were sent back for correction instead of entering the dataset, and the resulting its datasets were complete and gave the best mean accuracy change of all configurations (Table X). A faster but less disciplined model can thus be used safely, because its failures are caught. Across all LLMs, the synthetic datasets have 99.7% (GPT-4o mini) to 100.0% of their values filled.

Answer to RQ1. The control layer is what makes LLM output usable as a dataset: direct prompting never returned the requested size and lost whole batches to malformed responses, whereas delivers complete datasets of the requested size, records its cost and catches the failures of less disciplined models, for 1.9% more tokens. Within one LLM, no design choice moves fidelity or utility beyond the noise floor, except

TABLE VII

CONTROL-LAYER VARIANTS (RQ1), ALL WITH GPT-4O MINI. THE FIRST FIVE ROWS RUN THE SAME PIPELINE AND DEFINE THE NOISE FLOOR. SECONDS AND TOKENS ARE MEANS PER REPETITION. BOLD: BEST VALUE AMONG THE LLM CONFIGURATIONS; LOWER IS BETTER FOR C2ST, FOR THE DISTANCE OF THE ATTACK AUC FROM 0.5, AND FOR COST.

	Fidelity and privacy				Utility change (pp)		Cost per repetition	
Configuration	JSD	Corr.	C2ST	Attack AUC	Acc.	F1	Seconds	Tokens
Noise floor: five runs of the same pipeline								
Reference SYNTHAGENT	72.70	72.57	0.878	0.526	−1.19	−0.14	378.7	73,510
No type rules	72.92	72.93	0.876	0.534	−1.80	−0.04	321.6	72,188
No post-processing	72.43	70.83	0.883	**0.523**	−1.71	+0.28	251.5	72,235
No context join	72.52	71.18	0.880	0.531	−1.31	−0.18	**228.1**	72,576
Logged rerun	72.72	72.05	0.885	0.530	−1.32	−0.45	282.9	73,235
Design choices								
Dense context	72.87	70.88	0.878	0.525	−1.24	+0.07	238.8	72,600
Diverse context	**73.27**	**73.72**	0.872	0.544	−1.34	−0.13	271.1	73,757
LLM-chosen settings	72.53	70.10	0.886	0.526	−1.25	+0.11	310.2	115,245
Range enforcement	71.83	71.72	**0.870**	0.534	−1.13	−0.79	339.8	**71,915**

TABLE VIII

REFERENCE CONFIGURATION (GPT-4O MINI) PER DATASET, AVERAGED OVER TEN REPETITIONS AND ALL CLASSIFIERS (RQ2, RQ3). JSD AND CORRELATION ARE ON A 0–100 SCALE.

	Fidelity and privacy				Acc. (%)		Change (pp)	
Dataset	JSD	Corr.	C2ST	AUC	Real	Aug.	Acc.	F1
Breast	77.2	74.5	0.911	0.550	95.1	94.4	−0.66	−0.62
Diabetes	80.2	44.0	0.897	0.504	74.2	74.1	−0.03	+2.39
Heart dis.	65.2	76.6	0.893	0.507	56.6	53.4	−3.15	−0.97
Heart fail.	71.2	81.2	0.892	0.534	74.1	71.6	−2.53	−1.00
Liver	71.8	73.8	0.883	0.495	67.4	67.6	+0.19	+0.13
Parkinson's	70.6	85.3	0.789	0.567	84.9	84.0	−0.97	−0.77
Mean	72.7	72.6	0.878	0.526	75.4	74.2	−1.19	−0.14

~~a small fidelity gain from diverse context. The value of the control layer lies in reliability rather than in a higher score, which leads to the next question: how good are the records it delivers?~~

B. RQ2: Fidelity and Privacy

The generated records follow the real distributions reasonably well without revealing the real records, but they remain distinguishable from them (Table VIII~~reports the reference configuration per dataset. Marginal fidelity~~). Fidelity averages 72.7 out of 100, from 65.2 on the five-class heart-disease dataset, the furthest below its required sample size~~(ratio 42.1)~~, to 80.2 on diabetes, the closest to it~~(ratio 1.0)~~. The membership-inference attack stays near chance ~~, at~~ (AUC 0.526 on average~~and~~ , at most 0.567~~on any dataset, so the synthetic records do not reveal which real records were used; this~~), which is a diagnostic for one attack ~~, not~~ rather than a formal guarantee. The C2ST accuracy of 0.878 shows that a classifier can still separate real from synthetic records, so ~~we do not claim that~~ the synthetic records ~~can replace real ones~~ complement real ones rather than replace them.

~~*Answer to RQ2.* The synthetic records follow the real marginals reasonably well without reproducing the real records, but they remain distinguishable from real data. Whether records of this quality help a classifier is the next question.~~

C. RQ3: Utility for Classifiers Trained on Small Data

~~Any utility effect must be judged against how much classifiers~~ The generated records help the classifiers that fail on the minority class and slightly harm those that are already strong, so their effect depends on the dataset and the classifier. Classifiers trained on real data alone already vary ~~, so we measure that first~~ widely (Table IX)~~. The variation is large. On~~ : on the 195-record Parkinson's dataset, MLP accuracy is 72.3 ± 18.2% ~~,~~ while tree ensembles exceed 92%~~. On heart failure ,~~ , and on heart failure SVM and kNN reach ~~63.3% and 61.3% accuracy but~~ over 60% accuracy with an F1 of ~~only 0.0 and 0.7, because they almost never~~ almost zero because they rarely predict the minority class~~; accuracy alone would hide the very failure that SDG is meant to address.~~

~~Real-only baselines (RQ3) over ten 80/20 splits: mean in percent, standard deviation in small type. XGBoost on liver disease (–) failed in every split because the labels are coded {1, 2}.~~

~~Against this background, the effect of synthetic records depends on both the dataset and the classifier (Table VIII, Figure 5)~~. With the reference configuration, ~~accuracy improves in 18 of~~ F1 improves in 26 of the 59 classifier–dataset ~~cells, decreases in 38 and is unchanged in three; F1 improves in 26,~~ combinations and decreases in 31 ~~and is unchanged in two. The~~ (Figure 5); the largest gains occur where real-only training fails~~on the minority class:~~ , such as kNN on heart failure ~~, with a real-only F1 of 0.7%, gains 15.3~~ (+15.3 F1 points~~,~~) and MLP on Parkinson's ~~, the most unstable baseline, gains 8.2 accuracy points. Classifiers that are already strong lose slightly. After Holm correction over 59 tests per metric, no single cell is significant at 0.05.~~ (+8.2 accuracy points). No single combination is significant after Holm correction, and Figure 6 shows the variation between ~~splits: the accuracy change is consistently negative on the two heart datasets and centred on zero for liver disease, and diabetes is the only~~

dataset with a clearly positive divisions. On average, F1 change at unchanged accuracy.

Change from real-only to real-plus-synthetic training for the reference configuration (RQ3), per classifier and dataset. Hatched cells (n/a) were not evaluated. The same synthetic data can help one classifier and harm another.

Variation between splits for the reference configuration (RQ3). Points are splits, after averaging the paired changes over classifiers; diamonds are means and bars 95% t intervals.

Answer to RQ3. With is unchanged with GPT-4o mini , synthetic records help the classifiers that fail on the minority class and slightly harm the strong ones; on average they leave F1 unchanged (-0.14 pp) and lower accuracy (-1.19 pp). This is what information theorylead one to expectincreases by 1.09 pp with GPT-4.1, which is consistent with the bound set by information theory: synthetic records cannot carry more information about the task than the real records and the generator's own knowledge together [6], so large average gains are unlikely. If the LLM is the source of that information, a stronger LLM should matter, and so should the choice between an LLM and a generator fitted to the data. RQ4 examines both. .

D. RQ4: Comparison with Established Generators and LLMs

The choice of SYNTHAGENTgives the largest and most stable F1 gain among the generators compared, and the LLM is the only factor whose effect exceeds the noise floor. Among the LLMs (Table X, Figure 7). All three alternatives to GPT-4o mini exceed it in accuracy change and two in F1 change. , GPT-4.1 gives the highest marginal fidelity (75.72, 2.8 points above the floor) and the largest F1 gain ($+1.09$ pp); , and GPT-4.1 mini gives the best accuracy change (-0.23 pp) at the lowest generation time. The ; the three stronger LLMs preserve pairwise correlations correlations between features less well (62.8 –64.3, against at least 70.83 to 64.3, against 72.6 for GPT-4o mini), so better marginals come at the cost of weaker dependencies. Across six datasets these . These differences are not significant ($\chi^2_F(3) = 2.00$, across six datasets ($p = .572$ for accuracy; $\chi^2_F(3) = 1.80$, $p = .615$ for F1); the closest comparison, GPT-4.1 mini against GPT-4o mini in accuracy, differs by $+0.86$ pp with $p_{\text{Holm}} = .188$.

How does LLM-based synthesis compare with generators fitted to the data? Against the established generators (Table XIplaces with GPT-4o mini and GPT-4.1 next to SMOTE, CTGAN and TVAE. No generator), no method is best on every measure. CTGAN reaches the highest marginal fidelity (JSD fidelity (77.6), but its records are the easiest to detect (C2ST 0.936) and only 29.6% of them are valid(1.0% on Parkinson's). ; TVAE keeps 75.6% valid records and SMOTE 85.7%, although but only 14.4% on heart disease, where interpolation produces fractional values in integer-coded features. All five generators keep the membership attack near chance(0.503–0.532). Cost separates them sharply: SMOTE needs 0.015 seconds, TVAE and CTGAN 5.6 and 14.6 . The established generators are much cheaper, needing seconds of CPU time , and 73,000 to 81instead of about 80,000 tokensand several minutes of API time, with no training.

Average changes, however, hide how much each generator can harm an individual datasetThe decisive difference appears per dataset (Table XII). Synthetic records can harm a classifier when they misrepresent the real data, and in small data a large loss on one dataset matters more than a small gain on average . Table XII reports the average gain. SMOTE gains 10.91 pp of F1 change per dataset. SMOTE gives the largest gains where the scored class is rare ($+10.91$ pp on heart failure , $+4.11$ on diabetes) but loses 8.20 pp on liver disease, where the scored class is the majority. ; CTGAN loses on every dataset, by up to 10.65 pp. TVAE gives the best accuracy change on average but ; TVAE loses 2.86 pp of F1 on heart failure. SYNTHAGENT with GPT-4.1 has both the largest mean F1 combines the largest average gain ($+1.09$ pp) and with the smallest worst case : it never lowers the F1 of a dataset by more than 0.53 pp, while raising it by (-0.53 pp), while gaining 6.35 pp on heart failure. Across datasets , the differences between the five generators the differences are not significant ($\chi^2_F(4) = 6.80$, $p = .147$ for accuracy; $\chi^2_F(4) = 5.60$, $p = .231$ for F1; every Holm-adjusted $p \geq .25$); TVAE has the best accuracy rank (2.00), and the two configurations share the best F1 rank (2.50).

Mean F1 change per dataset (pp) over ten splits and all classifiers (RQ4). SA: . Worst: largest loss on any dataset.

Answer to RQ4. The LLM is the main lever on quality and utility. With GPT-4.1), gives the largest and most consistent F1 gain of the five generators, at a higher cost than the established generators and without a significant advantage across six datasets. so TVAE remains a cheap option for accuracy , and SMOTE for a known rare positive class.

VI. DISCUSSION AND LIMITATIONS

We set out The objective of this work was to determine whether a control layer SYNTHAGENT makes LLM-based SDG a reliable and useful option for small data, and the four answers combine into a clear picturehealthcare datasets. Reliability is achieved, and it is achieved by the control layer: the same LLM, prompted directly, never returned the dataset that was requesteddelivered the configured dataset, whereas SYNTHAGENT completes itby construction, records completed it, recorded its cost and caught the failures of a less disciplined model , at almost no extra cost (RQ1). Usefulness is real but bounded: the synthetic records are faithful without reproducing real ones , yet distinguishable from them revealing real ones (RQ2); they help , they help the classifiers that fail on the minority class (RQ3); , and with a stronger LLM they give the most consistent F1 largest and most stable gain among the generators compared (RQ4). No generator, however, improves utility significantly on all datasets. This every dataset, which is consistent with the data-processing inequality [6]: SDG cannot replace missing patients, and its benefit is bounded by the knowledge the generator brings. Among the generators we compared, the LLM is the only one that compared, only the

TABLE IX

REAL-ONLY BASELINES (RQ3) OVER TEN 80/20 DIVISIONS: MEAN IN PERCENT, STANDARD DEVIATION IN SMALL TYPE. XGBOOST ON LIVER DISEASE (–) FAILED IN EVERY DIVISION BECAUSE THE LABELS ARE CODED $\{1, 2\}$.

	Heart dis.		Breast		Heart fail.		Diabetes		Liver		Parkinson's	
Classifier	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1
RF	61.0 $\pm$4.4	57.0 $\pm$5.9	95.6 $\pm$1.8	95.0 $\pm$2.8	84.3 $\pm$3.3	76.8 $\pm$4.4	74.9 $\pm$2.8	60.4 $\pm$6.0	71.1 $\pm$3.5	81.4 $\pm$2.4	92.3 $\pm$3.2	94.8 $\pm$2.2
DT	53.4 $\pm$3.9	54.0 $\pm$5.0	93.3 $\pm$2.3	92.1 $\pm$4.0	79.2 $\pm$4.7	68.4 $\pm$9.2	71.1 $\pm$3.8	57.2 $\pm$6.4	66.0 $\pm$3.2	76.3 $\pm$3.1	85.1 $\pm$5.6	89.7 $\pm$3.8
MLP	56.6 $\pm$6.4	52.5 $\pm$7.6	94.7 $\pm$0.9	94.0 $\pm$2.1	48.3 $\pm$12.9	34.2 $\pm$24.7	68.4 $\pm$4.1	52.9 $\pm$12.0	68.4 $\pm$4.1	79.4 $\pm$3.6	72.3 $\pm$18.2	77.8 $\pm$26.1
SVM	54.8 $\pm$6.8	39.0 $\pm$7.9	93.3 $\pm$2.1	92.4 $\pm$3.4	63.3 $\pm$5.2	0.0 $\pm$0.0	75.6 $\pm$2.2	57.3 $\pm$4.7	70.7 $\pm$2.9	82.8 $\pm$2.0	81.0 $\pm$8.2	88.5 $\pm$5.2
LR	59.8 $\pm$5.6	56.5 $\pm$5.0	95.2 $\pm$2.0	94.6 $\pm$2.6	80.5 $\pm$3.6	68.9 $\pm$8.4	77.3 $\pm$2.0	62.8 $\pm$5.4	70.3 $\pm$4.0	81.1 $\pm$3.2	86.9 $\pm$5.6	91.3 $\pm$3.9
kNN	55.4 $\pm$6.3	44.4 $\pm$7.9	94.3 $\pm$1.8	93.4 $\pm$3.2	61.3 $\pm$4.1	0.7 $\pm$2.1	75.5 $\pm$1.8	60.0 $\pm$3.3	68.6 $\pm$4.2	79.7 $\pm$3.2	83.6 $\pm$5.6	89.5 $\pm$3.9
NB	55.7 $\pm$5.3	56.9 $\pm$5.0	93.9 $\pm$1.7	93.1 $\pm$3.2	75.0 $\pm$4.9	56.7 $\pm$13.1	76.4 $\pm$2.4	64.0 $\pm$4.9	55.9 $\pm$3.7	55.9 $\pm$3.8	71.5 $\pm$4.4	76.6 $\pm$4.0
GB	54.1 $\pm$5.5	52.8 $\pm$6.3	95.9 $\pm$2.2	95.4 $\pm$2.8	83.0 $\pm$4.9	75.4 $\pm$7.2	74.7 $\pm$3.0	61.7 $\pm$5.8	67.9 $\pm$3.3	78.7 $\pm$2.4	92.1 $\pm$4.0	94.7 $\pm$2.6
AdaBoost	60.8 $\pm$7.6	60.7 $\pm$7.3	97.1 $\pm$1.5	96.8 $\pm$1.7	82.5 $\pm$3.0	74.2 $\pm$6.5	74.9 $\pm$2.2	60.8 $\pm$4.7	67.4 $\pm$3.4	78.2 $\pm$3.3	91.3 $\pm$3.8	94.2 $\pm$2.5
XGBoost	54.3 $\pm$4.6	52.0 $\pm$5.1	97.2 $\pm$1.4	96.9 $\pm$1.8	83.5 $\pm$4.0	75.5 $\pm$7.0	72.7 $\pm$2.8	59.1 $\pm$6.2	–	–	93.3 $\pm$5.2	95.6 $\pm$3.2

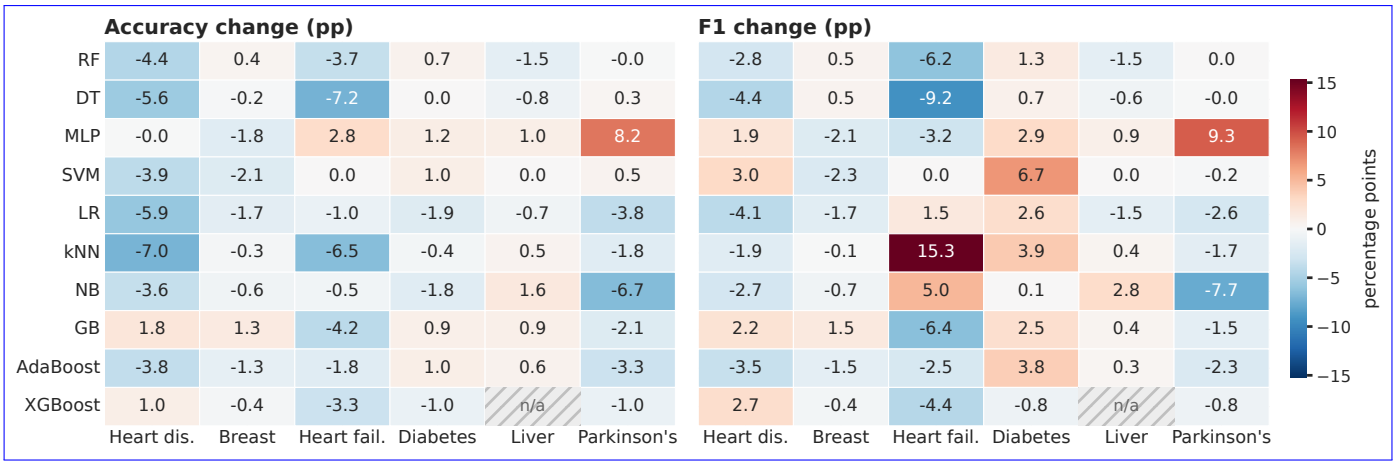


Fig. 5. Change from real-only to real-plus-synthetic training for the reference configuration (RQ3), per classifier and dataset. Hatched cells (n/a) were not evaluated. The same synthetic data can help one classifier and harm another.

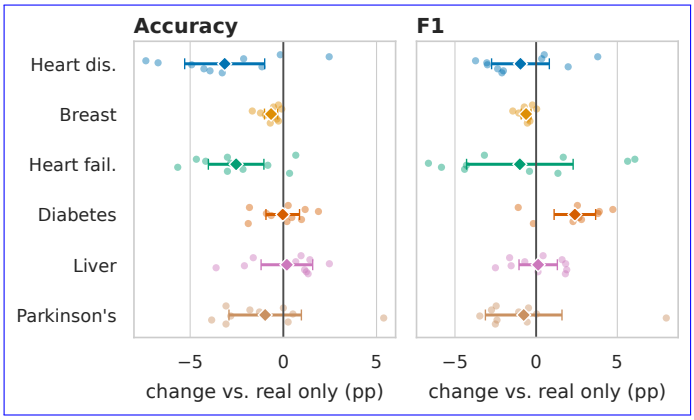


Fig. 6. Variation between divisions for the reference configuration (RQ3). Points are divisions, after averaging the paired changes over classifiers; diamonds are means and bars 95% t intervals.

LLM brings such knowledge, and the control layer is what makes it usable.

~~These results translate into four practical recommendations. Choose the LLM first, since it is the only factor whose effect~~

TABLE X

LLMS UNDER THE REFERENCE CONFIGURATION (RQ4). SECONDS AND TOKENS ARE MEANS PER REPETITION.

LLM	JSD	Corr.	ΔAcc.	ΔF1	Sec.	Tokens
GPT-4o mini	72.70	**72.57**	−1.19	−0.14	378.7	**73,510**
GPT-4.1 mini	73.60	62.82	−0.23	+0.92	**168.5**	72,069
GPT-4o	73.95	64.32	−0.50	+0.25	255.2	80,046
GPT-4.1	**75.72**	63.30	−0.44	+1.09	201.7	80,652

~~exceeds the variation between runs. Measure cost in tokens rather than time, which depends on the provider. Check~~ In practice, the choice of LLM matters most, cost is better measured in tokens than in time, and the effect of synthetic records ~~per classifier, on paired splits, before using them. And do not let the LLM choose its own settings without measuring the extra cost~~should be checked per classifier before they are used.

The conclusions are subject to several limitations. Only six datasets act as independent blocks, which limits the power of the tests, and the ~~splits~~ divisions are not stratified. The Parkinson's dataset contains several recordings per subject~~without a subject identifier~~, so recordings of one person can appear in

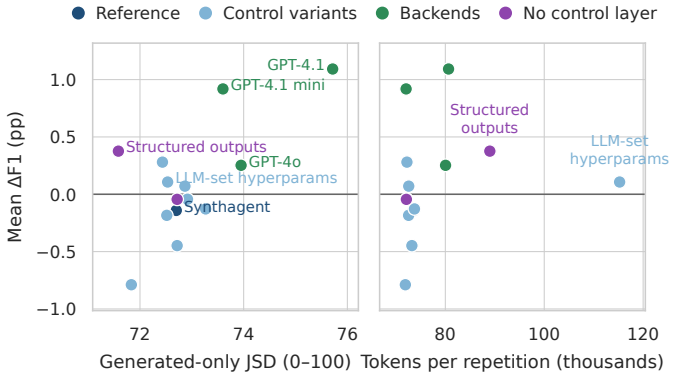


Fig. 7. Fidelity, F1 change and token cost of the LLM configurations (RQ1, RQ4). Control-layer variants cluster inside the noise floor; the LLMs spread along fidelity and F1; without the control layer, structured outputs cost the most tokens of the GPT-4o mini configurations except LLM-chosen settings.

TABLE XI
GENERATORS COMPARED ON THE SAME ~~SPLITS~~ DIVISIONS (RQ4). VALID: % OF RECORDS WITHIN THE OBSERVED RANGES, CATEGORIES AND INTEGER TYPES (NOT LOGGED FOR THE LLM RUNS, WHOSE TYPES ARE ENFORCED BY THE VALIDATOR). SECONDS: API TIME FOR SYNTHAGENT, CPU TIME OTHERWISE.

Generator	JSD	Corr.	C2ST	AUC	Valid	ΔAcc.	ΔF1	Sec.	Tok.
SYNTHAGENT, 4o mini	72.7	72.6	0.878	0.526	–	−1.19	−0.14	379	73.5k
SYNTHAGENT, 4.1	75.7	63.3	0.877	0.527	–	−0.44	+1.09	202	80.7k
SMOTE†	69.9	85.0	0.795	0.532	**85.7**	−3.08	+0.33	**0.01**	0
CTGAN	**77.6**	75.2	0.936	**0.503**	29.6	−1.40	−3.37	14.6	0
TVAE	71.9	**85.6**	0.868	0.515	75.6	−0.09	−0.21	5.6	0

†Emits only non-majority records, so its C2ST is not comparable.

both ~~partitions. Schema and value ranges are inferred once~~ sets. The columns and types are inferred from the full dataset ~~, which exposes test extremes to the profiler; train-only profiling would be stricter. JSD uses fixed histograms and is dominated by numerical features, and the membership attack is a single~~ rather than from the training set alone. The parameters in Table II were fixed and not tuned, and only two of them were varied, through the LLM-chosen settings variant. SYNTHAGENTdoes not filter synthetic records that are close to real ones: privacy is measured with one distance-based ~~adversary, so~~ attack, not enforced, and no result establishes differential privacy [39]. The SYNTHAGENT runs predate the counter of delivered records, so their size follows from the completion loop ~~of Algorithm 1~~ rather than from a logged count, and the validity measure was logged only for the established generators and direct prompting (78.6% valid records)~~, so it is not compared with . The correction paths~~ . The corrections were exercised only by GPT-4.1 mini~~; a stress test with weaker or locally served models and stricter rules is needed to measure how often each one is needed, and in the current implementation the output of the~~ , and records obtained in the last correction round ~~and the records obtained in top-up rounds~~ or in the completion loop are not validated again. Context selection is not class-stratified, so a rare class can be missing from a prompt. All LLMs come from one provider~~; GPT-4.1 nano and o4-mini were started but did not complete every dataset and are excluded.~~ , and CTGAN and TVAE use default settings~~, and tuned or diffusion-based generators may do better on some datasets. Finally, the sample-size ratios in Table IV rely on a default anticipated R^2_{CS} and are estimates~~.

TABLE XII
MEAN F1 CHANGE PER DATASET (PP) OVER TEN DIVISIONS AND ALL CLASSIFIERS (RQ4). SA: SYNTHAGENT. WORST: LARGEST LOSS ON ANY DATASET.

Generator	Heart d.	Breast	Heart f.	Diab.	Liver	Park.	Mean	Worst
SA, 4o mini	−0.97	−0.62	−1.00	+2.39	+0.13	−0.77	−0.14	−1.00
SA, 4.1	−0.44	−0.07	+6.35	+1.75	−0.50	−0.53	+1.09	−0.53
SMOTE	−1.50	−0.53	+10.91	+4.11	−8.20	−2.83	+0.33	−8.20
CTGAN	−2.30	−2.00	−10.65	−4.22	−0.65	−0.38	−3.37	−10.65
TVAE	−0.73	−0.88	−2.86	+0.89	+2.21	+0.10	−0.21	−2.86

Synthetic healthcare records can support development, teaching and software testing, but they must not inform diagnosis or policy without domain review, and near-chance performance against one ~~membership~~ attack does not justify releasing sensitive data. ~~To support verification, the~~ The implementation, the experiment runner and the script that regenerates every table and figure ~~from the stored results~~ are publicly available.[1]

VII. CONCLUSION

Small ~~clinical~~ healthcare datasets pose a dilemma for synthetic data generation~~. Generators~~ : generators fitted to the available records ~~cannot add the information those records lack~~can only recombine the patterns those records contain, whereas an LLM can ~~bring knowledge from pre-training~~ add domain knowledge but returns text rather than a dataset. SYNTHAGENT resolves the dilemma with a control layer that turns ~~LLM responses into complete, validated datasets of the requested size and records their~~ the responses of an LLM into complete synthetic datasets with the columns and types of the real data and a recorded cost, without training on the target data. On six ~~small~~ public healthcare datasets, ~~direct prompting never delivered the requested~~ neither direct prompting nor schema-constrained decoding ever delivered the configured dataset, while SYNTHAGENT ~~completes it by construction,~~ did at 1.9% more ~~tokens; with the larger, more recent~~ cost, and with the strongest LLM it gave the largest and most ~~consistent~~ stable F1 gain of five generators~~, although no generator was significantly better on every dataset~~. Because repeated runs of the same pipeline vary as much as most design choices, we recommend that SDG for small data be evaluated ~~against such a noise floor~~on repeated runs. Future work will evaluate locally served LLMs, class-aware context selection, privacy filters and the deployment across healthcare facilities outlined in Section III-E.

[1]TODO(authors): add the public repository URL required by the ICDE artifact policy.

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References

[1] N. Rieke, J. Hancox, W. Li, F. Milletar'i, H. R. Roth, S. Albarqouni, S. Bakas, M. N. Galtier, B. A. Landman, K. Maier-Hein, S. Ourselin, M. Sheller, R. M. Summers, A. Trask, D. Xu, M. Baust, and M. J. Cardoso, "The future of digital health with federated learning," *npj Digital Medicine*, vol. 3, no. 1, p. 119, 2020.

[2] R. D. Riley, K. I. E. Snell, J. Ensor, D. L. Burke, F. E. Harrell, K. G. M. Moons, and G. S. Collins, "Minimum sample size for developing a multivariable prediction model: Part II – binary and time-to-event outcomes," *Statistics in Medicine*, vol. 38, no. 7, pp. 1276–1296, 2019.

[3] Z. Qian, R. Davis, and M. van der Schaar, "Synthcity: A benchmark framework for diverse use cases of tabular synthetic data," in *Advances in Neural Information Processing Systems*, vol. 36, 2023.

[4] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002.

[5] L. Xu, M. Skoularidou, A. Cuesta-Infante, and K. Veeramachaneni, "Modeling tabular data using conditional GAN," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.

[6] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. Hoboken, NJ: Wiley-Interscience, 2006.

[7] V. Borisov, K. Sessler, T. Leemann, M. Pawelczyk, and G. Kasneci, "Language models are realistic tabular data generators," in *International Conference on Learning Representations*, 2023. [Online]. Available: https://openreview.net/forum?id=cEygmQNOeI

[8] N. Seedat, N. Huynh, B. van Breugel, and M. van der Schaar, "Curated LLM: Synergy of LLMs and data curation for tabular augmentation in low-data regimes," in *Proceedings of the 41st International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 235, 2024.

[9] R. Y. Wang and D. M. Strong, "Beyond accuracy: What data quality means to data consumers," *Journal of Management Information Systems*, vol. 12, no. 4, pp. 5–33, 1996.

[10] C. Batini, C. Cappiello, C. Francalanci, and A. Maurino, "Methodologies for data quality assessment and improvement," *ACM Computing Surveys*, vol. 41, no. 3, pp. 16:1–16:52, 2009.

[11] A. D. Lautrup, T. Hyrup, A. Zimek, and P. Schneider-Kamp, "SynthEval: A framework for detailed utility and privacy evaluation of tabular synthetic data," *Data Mining and Knowledge Discovery*, vol. 39, no. 1, p. 6, 2025.

[12] A. Kotelnikov, D. Baranchuk, I. Rubachev, and A. Babenko, "TabDDPM: Modelling tabular data with diffusion models," in *Proceedings of the 40th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 202, 2023, pp. 17 564–17 579.

[13] H. Zhang, J. Zhang, Z. Shen, B. Srinivasan, X. Qin, C. Faloutsos, H. Rangwala, and G. Karypis, "Mixed-type tabular data synthesis with score-based diffusion in latent space," in *International Conference on Learning Representations*, 2024. [Online]. Available: https://openreview.net/forum?id=4Ay23yeuz0

[14] A. V. Solatorio and O. Dupriez, "REaLTabFormer: Generating realistic relational and tabular data using transformers," *arXiv preprint arXiv:2302.02041*, 2023.

[15] J. Kim, T. Kim, and J. Choo, "EPIC: Effective prompting for imbalanced-class data synthesis in tabular data classification via large language models," in *Advances in Neural Information Processing Systems*, vol. 37, 2024.

[16] J. Zhang, G. Cormode, C. M. Procopiuc, D. Srivastava, and X. Xiao, "PrivBayes: Private data release via Bayesian networks," in *Proceedings of the ACM SIGMOD International Conference on Management of Data*, 2014.

[17] B. T. Willard and R. Louf, "Efficient guided generation for large language models," *arXiv preprint arXiv:2307.09702*, 2023.

[18] S. Geng, M. Josifoski, M. Peyrard, and R. West, "Grammar-constrained decoding for structured NLP tasks without finetuning," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023, pp. 10 932–10 952.

[19] C. Lee, J. Kim, and N. Park, "CoDi: Co-evolving contrastive diffusion models for mixed-type tabular synthesis," in *Proceedings of the 40th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 202, 2023, pp. 18 940–18 956.

[20] J. Fan, T. Liu, G. Li, J. Chen, Y. Shen, and X. Du, "Relational data synthesis using generative adversarial networks: A design space exploration," *Proceedings of the VLDB Endowment*, vol. 13, no. 11, pp. 1962–1975, 2020.

[21] A. Shankar, H. Brouwer, R. Hai, and L. Y. Chen, "SiloFuse: Cross-silo synthetic data generation with latent tabular diffusion models," in *Proceedings of the IEEE 40th International Conference on Data Engineering (ICDE)*, 2024, pp. 110–123.

[22] L. Chen, C. Wu, J. Li, S. Liu, G. Ye, and H. Chai, "TabLoft: Tabular data generation based on LLM with ordered features," in *Proceedings of the IEEE 42nd International Conference on Data Engineering (ICDE)*, 2026, pp. 1044–1056.

[23] J. Hyeong, J. Kim, N. Park, and S. Jajodia, "An empirical study on the membership inference attack against tabular data synthesis models," *arXiv preprint arXiv:2208.08114*, 2022.

[24] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao, "ReAct: Synergizing reasoning and acting in language models," in *International Conference on Learning Representations*, 2023. [Online]. Available: https://openreview.net/forum?id=WE_vluYUL-X

[25] L. Wang, C. Ma, X. Feng, Z. Zhang, H. Yang, J. Zhang, Z. Chen, J. Tang, X. Chen, Y. Lin, W. X. Zhao, Z. Wei, and J.-R. Wen, "A survey on large language model based autonomous agents," *Frontiers of Computer Science*, vol. 18, no. 6, p. 186345, 2024.

[26] A. Sanghi and J. R. Haritsa, "Synthetic data generation for enterprise DBMS," in *Proceedings of the IEEE 39th International Conference on Data Engineering (ICDE)*, 2023, pp. 3585–3588.

[27] C. Ge, S. Mohapatra, X. He, and I. F. Ilyas, "Kamino: Constraint-aware differentially private data synthesis," *Proceedings of the VLDB Endowment*, vol. 14, no. 10, pp. 1886–1899, 2021.

[28] X. Chu, I. F. Ilyas, and P. Papotti, "Holistic data cleaning: Putting violations into context," in *Proceedings of the IEEE 29th International Conference on Data Engineering (ICDE)*, 2013, pp. 458–469.

[29] J. Qin, S. Huang, Y. Wang, J. Zhu, Y. Zhang, Y. Miao, R. Mao, M. Onizuka, and C. Xiao, "BClean: A Bayesian data cleaning system," in *Proceedings of the IEEE 40th International Conference on Data Engineering (ICDE)*, 2024, pp. 3407–3420.

[30] A. Narayan, I. Chami, L. Orr, S. Arora, and C. Ré, "Can foundation models wrangle your data?" *Proceedings of the VLDB Endowment*, vol. 16, no. 4, pp. 738–746, 2022.

[31] J. Rorseth, P. Godfrey, L. Golab, D. Srivastava, and J. Szlichta, "Recovering structure in unstructured LLM outputs," in *Proceedings of the IEEE 42nd International Conference on Data Engineering (ICDE)*, 2026, pp. 4329–4330.

[32] J. Liang, C. Lei, X. Qin, J. Zhang, A. Katsifodimos, C. Faloutsos, and H. Rangwala, "Featpilot: Automatic feature augmentation on tabular data," in *Proceedings of the IEEE 41st International Conference on Data Engineering (ICDE)*, 2025, pp. 2148–2160.

[33] P. Li, X. Rao, J. Blase, Y. Zhang, X. Chu, and C. Zhang, "CleanML: A study for evaluating the impact of data cleaning on ML classification tasks," in *Proceedings of the IEEE 37th International Conference on Data Engineering (ICDE)*, 2021, pp. 13–24.

[34] S. Guha, F. A. Khan, J. Stoyanovich, and S. Schelter, "Automated data cleaning can hurt fairness in machine learning-based decision making," in *Proceedings of the IEEE 39th International Conference on Data Engineering (ICDE)*, 2023, pp. 3747–3754.

[35] D. Lopez-Paz and M. Oquab, "Revisiting classifier two-sample tests," in *International Conference on Learning Representations*, 2017. [Online]. Available: https://openreview.net/forum?id=SJkXfE5xx

[36] R. Shokri, M. Stronati, C. Song, and V. Shmatikov, "Membership inference attacks against machine learning models," in *2017 IEEE Symposium on Security and Privacy*, 2017, pp. 3–18.

[37] A. Machanavajjhala, J. Gehrke, D. Kifer, and M. Venkitasubramaniam, "ℓ-diversity: Privacy beyond k-anonymity," in *Proceedings of the 22nd International Conference on Data Engineering (ICDE)*, 2006, p. 24.

[38] N. Li, T. Li, and S. Venkatasubramanian, "t-closeness: Privacy beyond k-anonymity and ℓ-diversity," in *Proceedings of the IEEE 23rd International Conference on Data Engineering (ICDE)*, 2007, pp. 106–115.

[39] C. Dwork and A. Roth, *The Algorithmic Foundations of Differential Privacy*, ser. Foundations and Trends in Theoretical Computer Science. Now Publishers, 2014, vol. 9, no. 3–4.

[40] M. J. Sheller, B. Edwards, G. A. Reina, J. Martin, S. Pati, A. Kotrotsou, M. Milchenko, W. Xu, D. Marcus, R. R. Colen, and S. Bakas, "Federated learning in medicine: Facilitating multi-institutional collaborations without sharing patient data," *Scientific Reports*, vol. 10, p. 12598, 2020.

[41] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. Ag"uera y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, vol. 54. PMLR, 2017, pp. 1273–1282. [Online]. Available: https://proceedings.mlr.press/v54/mcmahan17a.html

[42] P. Kairouz, H. B. McMahan, B. Avent *et al.*, "Advances and open problems in federated learning," *Foundations and Trends in Machine Learning*, vol. 14, no. 1–2, pp. 1–210, 2021.

[43] K. Bonawitz, V. Ivanov, B. Kreuter, A. Marcedone, H. B. McMahan, S. Patel, D. Ramage, A. Segal, and K. Seth, "Practical secure aggregation for privacy-preserving machine learning," in *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security*, 2017, pp. 1175–1191.

[44] Q. Chen, Y. Chen, B. Yan, X. Jiang, X. Zhang, Y. Kang, T. Zhang, W. Huang, C. Gao, L. Fan, and Q. Yang, "Model trip: Enhancing privacy and fairness in model fusion across multi-federations for trustworthy global healthcare," in *Proceedings of the IEEE 40th International Conference on Data Engineering (ICDE)*, 2024.

[45] A. Janosi, W. Steinbrunn, M. Pfisterer, and R. Detrano, "Heart disease," UCI Machine Learning Repository, 1988.

[46] W. Wolberg, O. Mangasarian, N. Street, and W. Street, "Breast cancer wisconsin (diagnostic)," UCI Machine Learning Repository, 1993.

[47] "Heart failure clinical records," UCI Machine Learning Repository, 2020.

[48] B. V. Ramana, M. S. P. Babu, and N. B. Venkateswarlu, "ILPD (indian liver patient dataset)," UCI Machine Learning Repository, 2012.

[49] M. Little, "Parkinsons," UCI Machine Learning Repository, 2007.

[50] J. W. Smith, J. E. Everhart, W. C. Dickson, W. C. Knowler, and R. S. Johannes, "Using the ADAP learning algorithm to forecast the onset of diabetes mellitus," in *Proceedings of the Annual Symposium on Computer Application in Medical Care*, 1988, pp. 261–265.

[51] J. Demšar, "Statistical comparisons of classifiers over multiple data sets," *Journal of Machine Learning Research*, vol. 7, pp. 1–30, 2006.

[52] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.